An Explainable AI Model for IoMT Network Attack Detection

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Abstract

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1 Introduction

The Internet of Medical Things (IoMT) is a subset of the rapidly growing Internet of Things (IoT), specifically focused on devices commonly found in medical settings. IoMT facilitates the seamless gathering of information from various healthcare devices, which can encompass wearable devices and sensors used to monitor patients. The healthcare sector's rapid adoption of this technology is primarily a result of its promising ability to lower patient-related expenses; medical appointments can be less frequent due to remote patient monitoring. Machine learning algorithms can be used to process the vast data produced by IoMT, enabling advanced analytics such as patient risk predictions and anomaly detection. For example, one study combines data collected from IoMT devices with a Machine Learning model to accurately predict patients' risk of suffering from heart disease. The study found that key risk factors linked to heart disease can be identified, and patients can be treated sooner, potentially improving patient outcomes. As these systems become more common in medical settings, securing how we collect, process, and store this data is critical.

References